

Web Data Integration Project Abstract (Group 6)

XXXX XXXX[XXXXXXXX], XXXX XXXX[XXXXXXXX], XXXX
XXXX[XXXXXXXX], XXXX XXXX[XXXXXXXX] Lucia Bolduan^[2261625],
and Safa Shaghely^[2480414]

Abstract. This project considers three different book datasets to combine information from different sources in a common schema. The datasets are a Goodreads dataset from Kaggle, book data from DBpedia, and an Amazon Books dataset from Hugging Face. The datasets are expected to contain some of the same books, but they differ in their attributes, attribute names, and data formats. The goal is to identify matching books across the different sources and combine their information.

The three datasets contain between 20,981 and 52,478 records. The planned integrated schema contains common attributes such as title, authors, publisher, publication date, number of pages, language, and genres. Depending on the source, additional information such as ratings, number of ratings, price, series, and awards is available. Matching books will be identified using a combination of common attributes, especially title and authors. By integrating the three sources, information about the same book can be combined in a common schema.

1 Content

1.1 Use Case

The goal of this project is to integrate book information from three different data sources. The sources are expected to contain some of the same books. However, they provide different information about the books and use different attribute names and data formats.

The integration aims to identify matching books and combine their information in a common schema. The resulting dataset can contain information such as title, authors, publisher, publication date, number of pages, language, genres, ratings, and price.

1.2 Requirements Fulfillment

Datasets Three different datasets are used for this project: a Goodreads dataset from Kaggle, book data from DBpedia, and an Amazon Books dataset from Hugging Face. The Goodreads and Amazon datasets are provided in CSV format.

The DBpedia data was retrieved in XML format using a SPARQL query executed in Python. The SPARQL query returned approximately 20,000 results. The DBpedia dataset used in this project contains 20,981 entities.

Table 1 gives an overview of the three datasets, including their sources, formats, number of entities, and available attributes.

The project requirements are addressed as follows:

- **Three different datasets:** The project uses three different data sources: Goodreads, DBpedia, and Amazon Books. This requirement is fulfilled.
- **At least 2,500 entities in the integrated dataset:** The three input datasets contain 52,478 entities (Goodreads), 20,981 entities (DBpedia), and 40,000 entities (Amazon). All three datasets are also within the recommended range of 10,000 to 100,000 entities. Since the datasets have not yet been integrated, the exact number of entities in the integrated dataset cannot be determined at this stage. However, based on the size of the input datasets, the required minimum of 2,500 entities is expected to be reached.
- **At least 1,000 entities contained in at least two datasets:** Since the datasets have not yet been integrated and the books have not yet been matched across the datasets, the exact number of shared entities cannot be determined at this stage. The expected overlap is discussed in the section *Expected Entity Overlap*.
- **At least eight attributes in the integrated dataset:** The planned integrated schema contains 12 attributes and therefore meets this requirement. Books can be identified using a combination of at least two attributes. In particular, `title` and `authors` can be used for this purpose.
- **At least five attributes contained in at least two datasets:** Ten attributes of the planned integrated schema are available in at least two of the three input datasets. Seven attributes – `title`, `authors`, `publisher`, `publicationDate`, `pages`, `language`, and `genres` – are available in all three datasets. This requirement is therefore fulfilled.
- **At least one list attribute:** The planned integrated schema contains several list attributes, including `authors`, `genres`, and `awards`. This requirement is fulfilled.

The datasets also differ in their completeness. Attributes with more than 30% missing values are marked with MV (Missing Values) in Table 1. In the Goodreads dataset, this applies to `series`, `characters`, `edition`, `firstPublishDate`, `awards`, and `setting`. In the DBpedia dataset, `genres`, `language`, and `year` have more than 30% missing values.

Integrated Schema The attributes of the three datasets are mapped to a common schema (Table 2). Attributes that describe the same information but have different names in the input datasets are mapped to the same attribute. For example, `author` in Goodreads and `authors` in DBpedia and Amazon are mapped to the common attribute `authors`. Similarly, `publishDate`, `year`, and `publication_date` are mapped to `publicationDate`.

Table 1. Overview of the analyzed datasets.

Dataset	Source	Format	# Ent.	# Attr.	List of attributes
Goodreads "Best Books Ever"	Kaggle ¹	CSV	52,478	23	title, series (MV), author, rating, description, language, isbn, genres, characters (MV), bookFormat, edition (MV), pages, publisher, publishDate, firstPublishDate (MV), awards (MV), numRatings, ratingsByStars, likedPercent, setting (MV), bbeScore, bbeVotes, price
DBpedia	DBpedia ²	XML	20,981	8	authors, genres (MV), isbns, language (MV), pages, publisher, title, year (MV)
Amazon Reviews 2023	Hugging Face ³	CSV	40,000	11	title, subtitle, authors, publisher, publication_date, language, pages, genres, average_rating, ratings_count, price

¹ <https://www.kaggle.com/datasets/meetnaren/goodreads-best-books>² <https://dbpedia.org/sparql>³ <https://huggingface.co/datasets/McAuley-Lab/Amazon-Reviews-2023>

Table 2 shows the planned integrated schema and the corresponding attributes in the input datasets. The planned schema contains 12 attributes. Ten of these attributes are available in at least two datasets. The attributes **title**, **authors**, **publisher**, **publicationDate**, **pages**, **language**, and **genres** can be mapped to attributes from all three datasets.

The shared attributes can later be used to identify matching books across the datasets. For example, a combination of **title** and **authors** can be used for this purpose.

Some attributes differ in their representation and level of detail. For example, DBpedia provides only the publication year, while Goodreads and Amazon provide a publication date. Such differences have to be considered during the integration process.

The datasets also provide different information about books. Goodreads and Amazon contain rating information, while DBpedia does not. Goodreads provides additional information such as series and awards, while Amazon also provides subtitles. The integration can therefore combine information about a book from different sources.

Explanation The three datasets contain information about the same type of entities, but provide different information about them. For example, Goodreads and Amazon contain rating information, while DBpedia does not. Goodreads also provides information such as series and awards, while Amazon contains additional information such as subtitles. At the same time, several attributes,

Table 2. Attribute Intersection with Integrated Schema

Attribute name	Type	Datasets in which the attribute is found
title	string	GR (title), DB (title), AM (title)
authors	list<string>	GR (author), DB (authors), AM (authors)
publisher	string	GR (publisher), DB (publisher), AM (publisher)
publicationDate	date	GR (publishDate), DB (year), AM (publication_date)
pages	integer	GR (pages), DB (pages), AM (pages)
language	string	GR (language), DB (language), AM (language)
genres	list<string>	GR (genres), DB (genres), AM (genres)
rating	float	GR (rating), AM (average_rating)
ratingsCount	integer	GR (numRatings), AM (ratings_count)
price	float	GR (price), AM (price)
series	string	GR (series)
awards	list<string>	GR (awards)
<i>Identifier for the gold standard only (not used for matching):</i>		
isbn	string	GR (isbn), DB (isbns), AM (isbn13, isbn10)

Notes: One record per *work*. GR = Goodreads Best Books Ever, DB = DBpedia, AM = Amazon; source column names in brackets.

including title, authors, publisher, pages, language, and genres, are shared across all three datasets.

Therefore, integrating the datasets can provide more complete information about a book than using only one of the datasets. The shared attributes can also be used to identify books that occur in multiple datasets.

Expected Entity Overlap The exact number of books that occur in more than one dataset is not yet known, because the three datasets have not yet been matched. Since all three datasets contain book data and share several attributes, some overlap between the datasets is expected.